

# Class 14: Pathway Analysis

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## Table of contents

|                                                      |    |
|------------------------------------------------------|----|
| Section1. Differential Expression Analysis . . . . . | 1  |
| Running DESeq2 . . . . .                             | 5  |
| Volcano Plot . . . . .                               | 6  |
| Adding Gene Annotation . . . . .                     | 8  |
| Section 2. Pathway Analysis . . . . .                | 10 |
| Section 3. Gene Ontology (GO) . . . . .              | 18 |
| Section 4. Reactome Analysis . . . . .               | 19 |

## Section1. Differential Expression Analysis

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Loading required package: generics
```

```
Attaching package: 'generics'
```

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

```
IQR, mad, sd, var, xtabs
```

The following objects are masked from 'package:base':

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

```
findMatches
```

The following objects are masked from 'package:base':

```
expand.grid, I, unname
```

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,  
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,  
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,  
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,  
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,  
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,  
colWeightedMeans, colWeightedMedians, colWeightedSds,  
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,  
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,  
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,  
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,  
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,  
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,  
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,  
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with  
'browseVignettes()'. To cite Bioconductor, see  
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

# Import metadata and take a peek
colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
          condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
countData = read.csv(countFile, row.names=1)
head(countData)
```

```
          length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
          SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

Q. Complete the code below to remove the troublesome first column from countData

```
# Note we need to remove the odd first $length col
countData <- as.matrix(countData[,-1])
head(countData)
```

|                 | SRR493366 | SRR493367 | SRR493368 | SRR493369 | SRR493370 | SRR493371 |
|-----------------|-----------|-----------|-----------|-----------|-----------|-----------|
| ENSG00000186092 | 0         | 0         | 0         | 0         | 0         | 0         |
| ENSG00000279928 | 0         | 0         | 0         | 0         | 0         | 0         |
| ENSG00000279457 | 23        | 28        | 29        | 29        | 28        | 46        |
| ENSG00000278566 | 0         | 0         | 0         | 0         | 0         | 0         |
| ENSG00000273547 | 0         | 0         | 0         | 0         | 0         | 0         |
| ENSG00000187634 | 124       | 123       | 205       | 207       | 212       | 258       |

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```
# Filter count data where you have 0 read count across all samples.
countData = countData[ rowSums(countData) > 0, ]
head(countData)
```

|                 | SRR493366 | SRR493367 | SRR493368 | SRR493369 | SRR493370 | SRR493371 |
|-----------------|-----------|-----------|-----------|-----------|-----------|-----------|
| ENSG00000279457 | 23        | 28        | 29        | 29        | 28        | 46        |
| ENSG00000187634 | 124       | 123       | 205       | 207       | 212       | 258       |
| ENSG00000188976 | 1637      | 1831      | 2383      | 1226      | 1326      | 1504      |
| ENSG00000187961 | 120       | 153       | 180       | 236       | 255       | 357       |
| ENSG00000187583 | 24        | 48        | 65        | 44        | 48        | 64        |
| ENSG00000187642 | 4         | 9         | 16        | 14        | 16        | 16        |

## Running DESeq2

```
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
res <- results(dds)
summary(res)
```

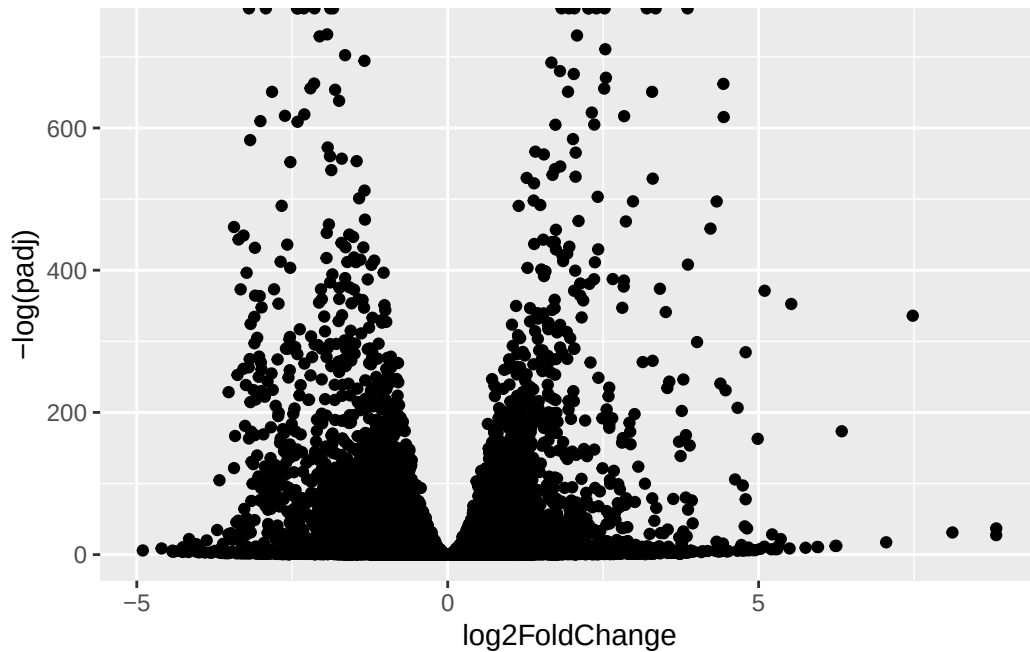
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

## Volcano Plot

```
library(ggplot2)

ggplot(res) +
  aes(x = log2FoldChange, y = -log(padj)) +
  geom_point()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
# Make a color vector for all genes
mycols <- rep("gray", nrow(res))

# Color blue the genes with fold change above 2
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

# Color gray those with adjusted p-value more than 0.01
mycols[ res$padj > 0.01 ] <- "gray"

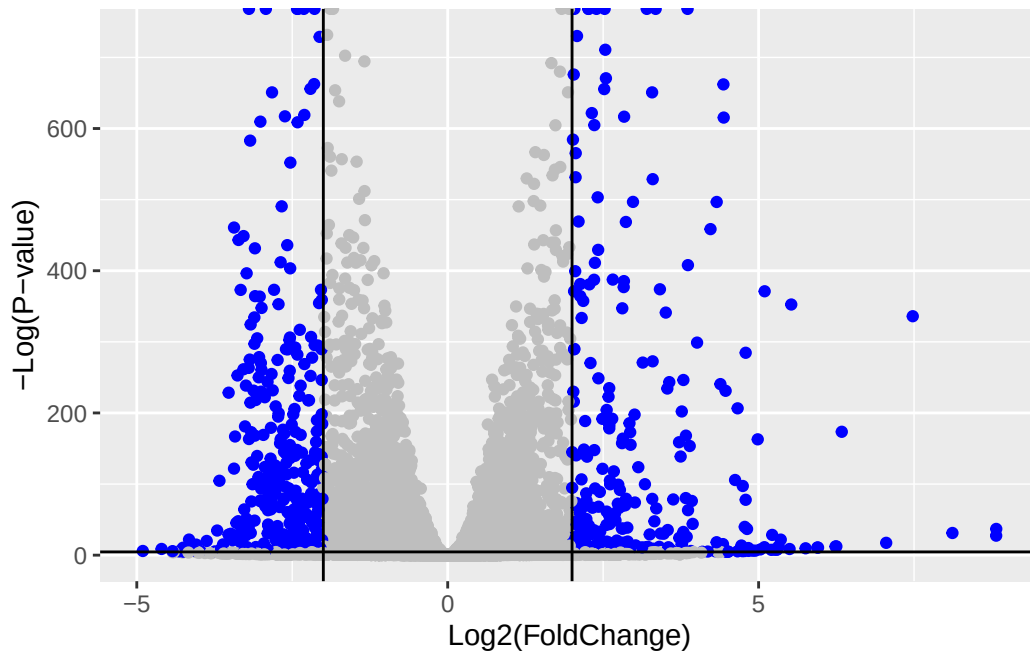
ggplot(res) +
```

```

aes(x = log2FoldChange, y = -log(padj)) +
geom_point(col = mycols) +
xlab("Log2(FoldChange)") +
ylab("-Log(P-value)") +
geom_vline(xintercept = c(-2,2)) +
geom_hline(yintercept = -log(0.01))

```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



## Adding Gene Annotation

Q. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```

library("AnnotationDbi")
library("org.Hs.eg.db")

```



```
res$symbol = mapIds(org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "SYMBOL",
  multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez = mapIds(org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "ENTREZID",
  multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name = mapIds(org.Hs.eg.db,
  keys = row.names(res),
  keytype = "ENSEMBL",
  column = "GENENAME",
  multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

|                 | baseMean    | log2FoldChange | lfcSE     | stat       | pvalue      |
|-----------------|-------------|----------------|-----------|------------|-------------|
|                 | <numeric>   | <numeric>      | <numeric> | <numeric>  | <numeric>   |
| ENSG00000279457 | 29.913579   | 0.1792571      | 0.3248215 | 0.551863   | 5.81042e-01 |
| ENSG00000187634 | 183.229650  | 0.4264571      | 0.1402658 | 3.040350   | 2.36304e-03 |
| ENSG00000188976 | 1651.188076 | -0.6927205     | 0.0548465 | -12.630156 | 1.43993e-36 |
| ENSG00000187961 | 209.637938  | 0.7297556      | 0.1318599 | 5.534326   | 3.12428e-08 |
| ENSG00000187583 | 47.255123   | 0.0405765      | 0.2718928 | 0.149237   | 8.81366e-01 |
| ENSG00000187642 | 11.979750   | 0.5428105      | 0.5215598 | 1.040744   | 2.97994e-01 |
| ENSG00000188290 | 108.922128  | 2.0570638      | 0.1969053 | 10.446970  | 1.51281e-25 |
| ENSG00000187608 | 350.716868  | 0.2573837      | 0.1027266 | 2.505522   | 1.22271e-02 |

|                 |             |             |             |                        |             |
|-----------------|-------------|-------------|-------------|------------------------|-------------|
| ENSG00000188157 | 9128.439422 | 0.3899088   | 0.0467164   | 8.346302               | 7.04333e-17 |
| ENSG00000237330 | 0.158192    | 0.7859552   | 4.0804729   | 0.192614               | 8.47261e-01 |
|                 | padj        | symbol      | entrez      |                        | name        |
|                 | <numeric>   | <character> | <character> |                        | <character> |
| ENSG00000279457 | 6.86555e-01 | NA          | NA          |                        | NA          |
| ENSG00000187634 | 5.15718e-03 | SAMD11      | 148398      | sterile alpha motif    | ..          |
| ENSG00000188976 | 1.76553e-35 | NOC2L       | 26155       | NOC2 like nucleolar    | ..          |
| ENSG00000187961 | 1.13413e-07 | KLHL17      | 339451      | kelch like family me.. |             |
| ENSG00000187583 | 9.19031e-01 | PLEKHN1     | 84069       | pleckstrin homology    | ..          |
| ENSG00000187642 | 4.03379e-01 | PERM1       | 84808       | PPARGC1 and ESRR       | ind..       |
| ENSG00000188290 | 1.30538e-24 | HES4        | 57801       | hes family bHLH tran.. |             |
| ENSG00000187608 | 2.37452e-02 | ISG15       | 9636        | ISG15 ubiquitin like.. |             |
| ENSG00000188157 | 4.21970e-16 | AGRN        | 375790      |                        | agrin       |
| ENSG00000237330 | NA          | RNF223      | 401934      | ring finger protein    | ..          |

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res <- res[order(res$padj), ]
write.csv(res, file = "deseq_results.csv")
```

## Section 2. Pathway Analysis

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
```

```
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```
foldchanges <- as.numeric(res$log2FoldChange)
names(foldchanges) <- res$entrez

head(foldchanges)
```

```
      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
attributes(keggres)
```

```
$names
[1] "greater" "less" "stats"
```

```
# Look at the first few down (less) pathways
head(keggres$less)
```

|                                       | p.geomean    | stat.mean | p.val        |
|---------------------------------------|--------------|-----------|--------------|
| hsa04110 Cell cycle                   | 8.995727e-06 | -4.378644 | 8.995727e-06 |
| hsa03030 DNA replication              | 9.424076e-05 | -3.951803 | 9.424076e-05 |
| hsa03013 RNA transport                | 1.375901e-03 | -3.028500 | 1.375901e-03 |
| hsa03440 Homologous recombination     | 3.066756e-03 | -2.852899 | 3.066756e-03 |
| hsa04114 Oocyte meiosis               | 3.784520e-03 | -2.698128 | 3.784520e-03 |
| hsa00010 Glycolysis / Gluconeogenesis | 8.961413e-03 | -2.405398 | 8.961413e-03 |
|                                       | q.val        | set.size  | exp1         |
| hsa04110 Cell cycle                   | 0.001448312  | 121       | 8.995727e-06 |

|                                       |             |     |              |
|---------------------------------------|-------------|-----|--------------|
| hsa03030 DNA replication              | 0.007586381 | 36  | 9.424076e-05 |
| hsa03013 RNA transport                | 0.073840037 | 144 | 1.375901e-03 |
| hsa03440 Homologous recombination     | 0.121861535 | 28  | 3.066756e-03 |
| hsa04114 Oocyte meiosis               | 0.121861535 | 102 | 3.784520e-03 |
| hsa00010 Glycolysis / Gluconeogenesis | 0.212222694 | 53  | 8.961413e-03 |

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14

Info: Writing image file hsa04110.pathview.png

```
# A different PDF based output of the same data
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"
```

Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14

Info: Writing image file hsa04110.pathview.pdf

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14

Info: Writing image file hsa04330.pathview.png

Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
# Focus on top 5 down-regulated pathways
keggrespathways <- rownames(keggres$less)[1:5]

# Extract the 8 character long IDs
keggresids <- substr(keggrespathways, start = 1, stop = 8)
keggresids
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

```
# Make pathview figures
pathview(gene.data = foldchanges, pathway.id = keggresids, species = "hsa")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14
```

```
Info: Writing image file hsa04110.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14
```

```
Info: Writing image file hsa03030.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14
```

```
Info: Writing image file hsa03013.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/parisa/Desktop/BIMM 143/Class14
```

```
Info: Writing image file hsa03440.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

Info: Writing image file hsa04114.pathview.png







### Section 3. Gene Ontology (GO)

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```

\$greater

|                                           | p.geomean    | stat.mean | p.val        |
|-------------------------------------------|--------------|-----------|--------------|
| G0:0007156 homophilic cell adhesion       | 8.519724e-05 | 3.824205  | 8.519724e-05 |
| G0:0002009 morphogenesis of an epithelium | 1.396681e-04 | 3.653886  | 1.396681e-04 |
| G0:0048729 tissue morphogenesis           | 1.432451e-04 | 3.643242  | 1.432451e-04 |
| G0:0007610 behavior                       | 1.925222e-04 | 3.565432  | 1.925222e-04 |
| G0:0060562 epithelial tube morphogenesis  | 5.932837e-04 | 3.261376  | 5.932837e-04 |
| G0:0035295 tube development               | 5.953254e-04 | 3.253665  | 5.953254e-04 |
|                                           | q.val        | set.size  | expl         |
| G0:0007156 homophilic cell adhesion       | 0.1951953    | 113       | 8.519724e-05 |
| G0:0002009 morphogenesis of an epithelium | 0.1951953    | 339       | 1.396681e-04 |
| G0:0048729 tissue morphogenesis           | 0.1951953    | 424       | 1.432451e-04 |
| G0:0007610 behavior                       | 0.1967577    | 426       | 1.925222e-04 |
| G0:0060562 epithelial tube morphogenesis  | 0.3565320    | 257       | 5.932837e-04 |
| G0:0035295 tube development               | 0.3565320    | 391       | 5.953254e-04 |

\$less

|                                          | p.geomean    | stat.mean | p.val        |
|------------------------------------------|--------------|-----------|--------------|
| G0:0048285 organelle fission             | 1.536227e-15 | -8.063910 | 1.536227e-15 |
| G0:0000280 nuclear division              | 4.286961e-15 | -7.939217 | 4.286961e-15 |
| G0:0007067 mitosis                       | 4.286961e-15 | -7.939217 | 4.286961e-15 |
| G0:0000087 M phase of mitotic cell cycle | 1.169934e-14 | -7.797496 | 1.169934e-14 |
| G0:0007059 chromosome segregation        | 2.028624e-11 | -6.878340 | 2.028624e-11 |
| G0:0000236 mitotic prometaphase          | 1.729553e-10 | -6.695966 | 1.729553e-10 |
|                                          | q.val        | set.size  | expl         |
| G0:0048285 organelle fission             | 5.841698e-12 | 376       | 1.536227e-15 |
| G0:0000280 nuclear division              | 5.841698e-12 | 352       | 4.286961e-15 |
| G0:0007067 mitosis                       | 5.841698e-12 | 352       | 4.286961e-15 |
| G0:0000087 M phase of mitotic cell cycle | 1.195672e-11 | 362       | 1.169934e-14 |

|            |                        |              |     |              |
|------------|------------------------|--------------|-----|--------------|
| G0:0007059 | chromosome segregation | 1.658603e-08 | 142 | 2.028624e-11 |
| G0:0000236 | mitotic prometaphase   | 1.178402e-07 | 84  | 1.729553e-10 |

\$stats

|            |                                | stat.mean | exp1     |
|------------|--------------------------------|-----------|----------|
| G0:0007156 | homophilic cell adhesion       | 3.824205  | 3.824205 |
| G0:0002009 | morphogenesis of an epithelium | 3.653886  | 3.653886 |
| G0:0048729 | tissue morphogenesis           | 3.643242  | 3.643242 |
| G0:0007610 | behavior                       | 3.565432  | 3.565432 |
| G0:0060562 | epithelial tube morphogenesis  | 3.261376  | 3.261376 |
| G0:0035295 | tube development               | 3.253665  | 3.253665 |

## Section 4. Reactome Analysis

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway with the most significant “Entities p-value” was the Cell Cycle Pathway. The significant pathways were similar to the previous KEGG results because both methods used the same gene expression data and found pathways related to cell division and the cell cycle. Some differences can happen because Reactome and KEGG use different databases, organize pathways differently, and use different statistical methods to calculate significance.